

**SAMPLE - NOT ISSUED.** No Engagement Terms are incorporated and no commitment of any kind is made. Issued reports carry the governing terms version here.

## SIRPY Verified — Tier 1 Report

*Raw computational output with the full honesty floor. This tier makes no formal certification claims; it reports what was measured.*

### Order identity

| Field                  | Value                                           |
|------------------------|-------------------------------------------------|
| Client                 | Acme Labs                                       |
| Order id               | LQI-2026-Troesch-15-2026-08-29-T1               |
| Reference              | LQI-2026-Troesch-15                             |
| Problem                | A plasma-confinement model.                     |
| Stated equation & data | $y'' = 15 \sinh(15y)$ , $y(0) = 0$ , $y(1) = 1$ |
| Ordered (date in)      | 2026-08-29                                      |
| Delivered (date out)   | 2026-08-29                                      |
| Service tier           | Tier 1: Verified                                |

### Run identity and finding

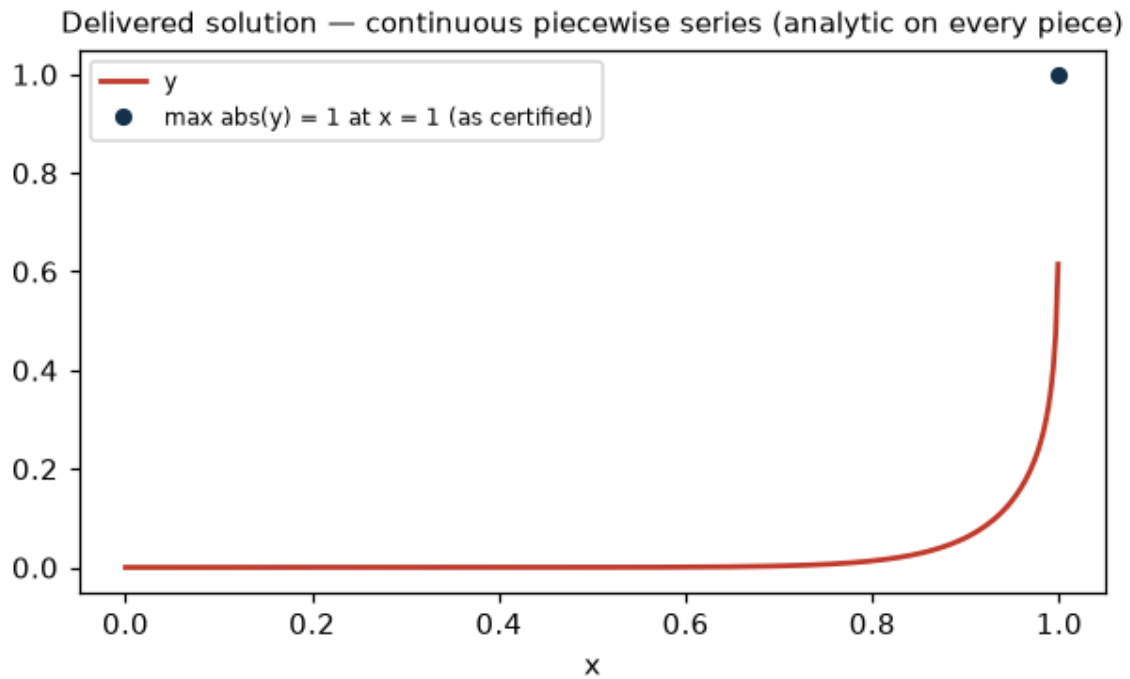
- Class: ODE boundary-value problem | Engine: SIRPY native — piecewise power series (35 pieces) | Generated by SIRPY 0.59.15
- FINDING: solved+verified
- max residual (measured): 1.588e-03 at  $x = 0.999995$
- artifact coverage:  $[0, 1]$ ; pieces solved 35, delivered 35
- artifact: LQI-2026-Troesch-15-T1.artifact.json
- sha256: 60a7f281 6691f97a eade4f2b 82aa76d3 3fec944a 330531c2 3c02bbe5 7e026df3
- reproduction: the headline numbers above are RECOMPUTABLE from this artifact and the stated equation using only Python and sha256sum -- substitute the delivered coefficients into the equation and compare. This is reproduction of our measurements, not agreement with a third-party solver: an independent solver may be compared against the delivered values, and any disagreement is a finding to report, not a covered reproducibility failure. Scope: the artifact coverage above.

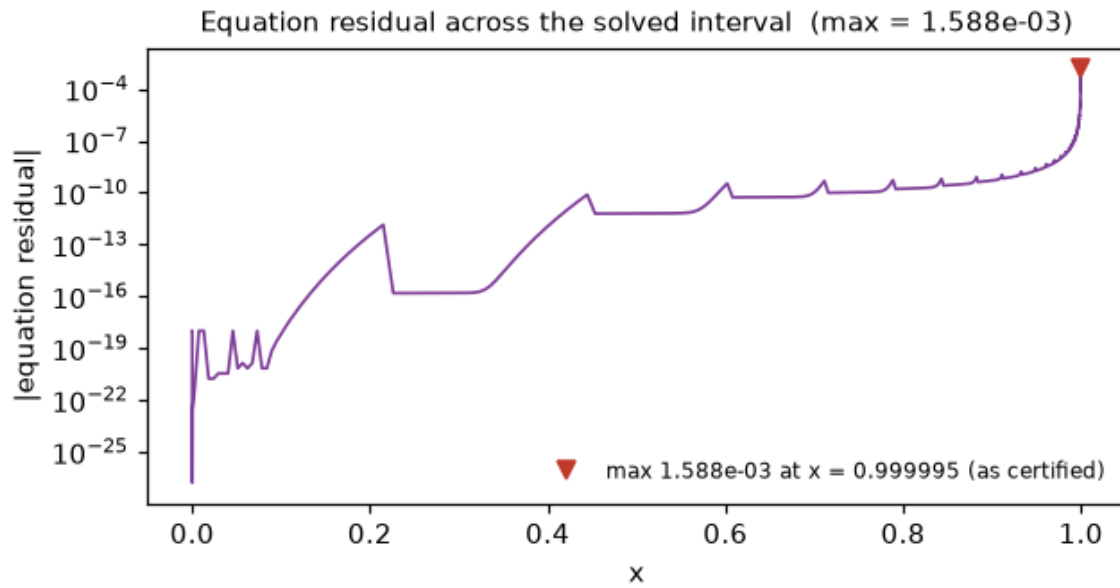
### Verify facts (measured on this run)

| Item                  | Value                                                        |
|-----------------------|--------------------------------------------------------------|
| checked               | the equation (residual by exact coefficient differentiation) |
| checked               | boundary data                                                |
| max equation residual | 0.00158845                                                   |

| Item                      | Value       |
|---------------------------|-------------|
| at                        | 0.999995    |
| boundary/initial residual | 5.84529e-10 |
| max interface jump        | 2.3487e-12  |
| interface nodes           | 34          |
| n_segments                | 35          |

## Figures





## Result

### PASS

Artifact: LQI-2026-Troesch-15-T1.artifact.json — sha256

60a7f2816691f97aeade4f2b82aa76d33fec944a330531c23c02bbe57e026df3. Every number above is reproducible from that file and the stated equation using Python and sha256sum alone.

Issued by LQI. The order manifest (ORDER\_AND\_DELIVERY.txt) is included in this delivery package alongside this document and the artifact; the terms of service applicable to this order are those supplied with your order confirmation and named by version in the delivery manifest.